

Vicky Bilbily

Human-centred software engineer with strong product sensibilities.

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website: vickybilbily.dev

WORK EXPERIENCE

Product Engineer, Educational Games

Aug 2023 – Present

Creatubbles

Remote, Canada

- Designed and built ~50% of the educational content for *Code Missions*, deployed to after-school STEM programs (Code Ninjas), private and public schools, with thousands of missions completed by students to date.
- Led end-to-end development of *SuperCode Saga*, an educational game series teaching CS fundamentals to K–12 students – translated learning goals into technical deliverables in a fast-paced, cross-functional startup environment, coordinating implementation across a remote team with evolving scope.
- Taught live coding classes, turning firsthand observation of how kids learn into both product and curriculum decisions – breaking complex CS concepts into structured, sequenced units grounded in real user behaviour.

Gameplay Engineer

Jun 2020 – May 2023

Behaviour Interactive

Montréal, Canada

- Built data-driven gameplay systems across the full-stack for the *Jurassic World: Primal Ops* mobile game, shipped to 100K+ players. Created major frontend UI flows with C# in Unity Engine, and developed backend business logic, player data modeling, and REST API endpoints in Javascript using Node and Express.
- Developed networked gameplay features for an unannounced PC/console title with C++ in Unreal Engine 5, ensuring performant client-server synchronization.
- Built intuitive in-engine tools for designers across both projects, balancing usability with technical constraints.
- Supported post-launch live operations, shipping fixes and new features while maintaining stability in production.

Graduate Researcher, Human-Computer Interaction

Sep 2018 – May 2020

University of Toronto

Toronto, Canada

- Published first-author research at UIST 2021 (top-tier ACM HCI conference): a direct-manipulation interface bridging digital maps and calendars, enabling flexible, constraint-based personal scheduling through spatial and temporal reasoning.
- Designed and built interactive prototypes in Swift (iOS) and TypeScript / React. Developed and conducted usability studies, synthesizing findings into academic writing and UX recommendations.
- Teaching assistant for introductory computer science and HCI courses.

Software Engineering Intern – Autodesk

May 2017 – Aug 2017

Software Engineering Intern – CIBC (Canadian Imperial Bank of Commerce)

Jan 2017 – Apr 2017

Research Intern – University of Toronto

May 2016 – Aug 2016

EDUCATION

Master of Science, Computer Science (Human-Computer Interaction)

Sep 2018 – Jan 2020

University of Toronto

Toronto, Canada

Bachelor of Engineering, Software Engineering

Sep 2013 – Apr 2018

McMaster University

Hamilton, Canada

SKILLS

Programming Languages: Javascript, Typescript, C#, C++, Swift, Lua, Python

Web Development: React, Node.js, Express.js, Mocha, REST APIs, HTML5, CSS3, Canvas, Git, NoSQL, Cursor AI

Design & Research: Figma, UX Design, UX Research, Usability Testing

Interactive & Game Development: Unity Engine, Unreal Engine, Processing, Canvas, Perforce

Spoken Languages: English (native), French (conversational)